

Comparative Analysis of LLM-Generated Event Timeline Summarization for Legal Investigations

Qiang Mao

Data & Technology
Ankura Consulting Group, LLC
Washington, D.C. USA
qiang.mao@ankura.com

Adam Dabrowski

Data & Technology
Ankura Consulting Group, LLC
Washington, D.C. USA
adam.dabrowski@ankura.com

Fusheng Wei

Data & Technology
Ankura Consulting Group, LLC
Washington, D.C. USA
fusheng.wei@ankura.com

Eric Olson

Data & Technology
Ankura Consulting Group, LLC
Washington, D.C. USA
eric.olson@ankura.com

Robert Neary

Data & Technology
Ankura Consulting Group, LLC
Washington, D.C. USA
robert.neary@ankura.com

Jingchao Yang

Data & Technology
Ankura Consulting Group, LLC
Washington, D.C. USA
jingchao.yang@ankura.com

Han Qin

Data & Technology
Ankura Consulting Group, LLC
Washington, D.C. USA
han.qin@ankura.com

Nathaniel Huber-Fliflet

Data & Technology
Ankura Consulting Group, LLC
Washington, D.C. USA
nathaniel.huber-fliflet@ankura.com

Abstract— This paper presents a comparative study evaluating the performance of Large Language Models (LLMs) in generating timeline summaries from construction delay documents. We assessed seven open-source LLMs and two commercial chatbots (ChatGPT and Claude) on their ability to extract, organize, and summarize delay events from twenty-one carefully curated synthetic snippets of text. The evaluation framework combined automatic metrics (BERTScore and ROUGE scores) with expert human assessment across four dimensions: event description accuracy, date accuracy, event capture completeness, and language quality.

Results demonstrate that while commercial solutions, particularly Claude, achieved superior performance, several open-source alternatives showed comparable capabilities. Notably, Llama-3.1-70B-Instruct showed robust performance in event capture and source tracking, while Llama-3.1-8B-Instruct offered efficient processing with balanced performance among smaller models. A critical finding was the widespread challenge in temporal information processing, with only Claude achieving complete accuracy in date extraction and event association. The study's findings suggest that open-source LLMs can serve as practical tools for construction document analysis, although model selection is a critical consideration based on specific accuracy and efficiency requirements, and resource constraints.

Keywords— LLMs, GenAI, Construction disputes, BERTScore, Expert Review, NLP

I. INTRODUCTION

Construction disputes are complex litigation matters that involve many key stakeholders across the course of the project. Construction projects can take years to complete and involve contributions from many parties to facilitate the project's completion. Lengthy original timelines coupled with delays

that organically arise during a project can extend these projects far beyond their estimated completion dates. During this period, millions of communications and documents are transmitted and collected by various parties, and in case of litigation, a relatively small number of relevant documents sit within the millions. As project timelines continue expanding, the document volumes increase and make it correspondingly harder to find relevant documents. Understanding how project delays can have cascading impacts over time and how they affect different stakeholders is critical in construction dispute litigation.

Creating a detailed, cohesive timeline of delay events on a construction matter using relevant documents and communications is crucial and can take dozens to hundreds of hours when done manually. This cumbersome manual review process quickly becomes excessively costly. In construction litigations, quickly finding the supporting documentation in a delay dispute is critical to the success of a matter. Accurately identifying and categorizing likely delay-related documents into their delay event categories establishes a set of prioritized and substantively organized documents for review.

The authors of this paper have done extensive work creating a predictive model trained on small text snippets of delay-related text. A text snippet is a small passage of words within a document usually ranging from 50 to 200 words [1]. In construction litigation, a document can be considered a delay event related document when one or more of its text snippets describe a delay event. This model has shown exemplary performance and its deployment on multiple real-world construction dispute matters, saving clients millions of dollars

in review costs. An extended feature of this delay model extracts delay-related snippets of text from documents and readily presents the key delay-related information. Using the captured snippets, the authors have also created an automated categorization process that organizes delay related snippets, and therefore the original documents, into defined, specific categories using proprietary taxonomies.

These collaborators have also conducted considerable research into integrating LLMs into legal domain workflows. As industries seek innovation and embrace advanced technologies, like LLMs, the legal industry sits at the forefront of this pursuit, given its widespread practice of applying predictive modeling to augment a manual document review and the text classification process. As LLMs continue evolving into state-of-the-art deep learning methods, exploring their integration and applications in legal disputes becomes imperative to continue streamlining dispute workflows.

Recent advances in LLMs have revolutionized natural language processing tasks, demonstrating remarkable capabilities in text generation, summarization, and information extraction and have surpassed deep learning models as the state-of-the-art architecture in every NLP [2, 3]. These models, trained on vast amounts of text data, can understand context, generate coherent responses, and follow complex instructions [4]. The emergence of open-source LLMs has made these capabilities more accessible, offering alternatives to commercial solutions for specialized applications [5].

Text generation and summarization capabilities of LLMs have shown promise in document analysis tasks [6]. When properly prompted, these models can extract relevant information, maintain temporal consistency, and present information in structured formats [7]. The application of LLMs to specialized document analysis tasks represents a promising direction for improving efficiency in document review processes [14]. This capability is especially valuable in complex domains like construction litigation, where accurate information extraction and timeline generation are crucial. Using this LLM capability in conjunction with the highly relevant extracted snippets of text provides a pathway to enrich the construction dispute document review workflow by (i) applying predictive modeling to classify relevant documents, (ii) extract meaningful snippets of text, (iii) organizing snippets and documents into event categories, and (iv) using an LLM to summarize and create a timeline for the critical events.

However, careful evaluation of LLM's performance, particularly in handling temporal information and maintaining factual accuracy, is essential for practical implementation [15]. The evaluation of LLM-generated content requires both automated metrics and human assessment. BERTScore, which leverages contextual embeddings from BERT models, provides a semantic similarity measure that correlates well with human judgment [8]. This metric offers advantages over traditional lexical-based measures by capturing semantic meaning rather than exact word matches [9]. ROUGE scores complement this

by measuring n-gram overlap between generated and reference texts, providing insights into content coverage and precision [10].

Expert review remains crucial for evaluating specialized content, particularly in technical domains [11]. To ensure reliability in human evaluation, inter-rater agreement measures such as Krippendorff's alpha are employed to quantify consistency across multiple expert assessments [12]. These measures help validate the reliability of human evaluations, especially when dealing with subjective aspects of text quality and accuracy [13].

This study contributes to this growing body of research by evaluating both commercial and open-source LLMs in the specific context of construction delay documentation analysis. In addition to the predictive modeling and categorization workflow, integrating an automated timeline generation process would drastically reduce review time and costs. There is potential for attorneys to use this complete process and access important timeline events and information without having to review any documents. This process leverages construction dispute litigation due to its complexity, length, and stakeholder involvement, but the overarching goal is to make this process industry agnostic and applicable to any litigation subject.

II. METHODOLOGY

This study evaluated nine LLMs, including seven open-source models and two commercial chatbots (ChatGPT and Claude) as reference benchmarks. All models were tested using identical prompts and configuration settings to ensure uniform comparison. The prompt for each model included the following details for summary generation:

- A 100-word concise overview of delay events
- Categorized timeline of events:
 - Dated events (with temporal information)
 - Undated events
- Source attribution (snippet IDs) for each event

Each model performed five independent generations to account for output variability. Multiple generations enabled calculating mean performance metrics and standard deviations.

A reference summary used to evaluate LLM outputs was manually created by experienced construction dispute analysts. Creating the reference summary is fully detailed in the Experiments section. The experts independently created individual summaries, and then produced a final consolidated version. Having multiple domain experts create the reference summary served multiple purposes:

1. Minimized individual bias and subjective interpretation of events.
2. Ensured comprehensive capture of critical delay events through multiple expert perspectives.
3. Validated the temporal sequencing of events.

4. Verified the technical accuracy of construction-specific terminology and causality relationships.
5. Guaranteed adherence to the required format while maintaining professional language standards.

The final reference summary underwent additional review to ensure it met all prompt requirements, including the 100-word limit for the overview section, proper categorization of dated and undated events, and accurate source attribution. This rigorous process established a high-quality benchmark against which the LLM-generated summaries could be reliably evaluated using automated metrics such as ROUGE scores and BERTScore.

A. Empirical Metrics

The following metrics were calculated for each output iteration and provide an empirical evaluation of the LLMs ability to effectively answer the prompt via the output content, and time needed to create the output.

1) ROUGE Scores

ROUGE (Recall-Oriented Understudy for Gisting Evaluation) metrics assess the quality of generated summaries by measuring their overlap with the reference summary. ROUGE-N (N=1,2) and ROUGE-L were computed using the human-created reference summary and the mean value and standard deviation were calculated after five iterations per model. The difference between each ROUGE calculation is:

- ROUGE-N: Measures n-gram overlap between generated and reference summary.
- ROUGE-L: Evaluates the longest common subsequence, capturing sentence-level structure.

2) BERTScore

BERTScore leverages pre-trained BERT embeddings to compute similarity between generated and reference texts, offering a more semantically nuanced evaluation than traditional lexical matching. This metric accounts for:

- Token-level semantic similarity
- Contextual understanding
- Soft matching of related terms

The BERTScore evaluation provided semantic similarity assessment through precision, recall, and F1 scores, utilizing a standard pre-trained BERT model and averaging results across multiple runs.

3) Time Efficiency

Execution time for each model was recorded across the five runs. To assess computational efficiency, we measured generation time for each run, calculating mean and standard deviation while excluding initialization time to ensure accurate performance metrics and providing insights into practical deployment considerations.

B. Human Evaluation

A human review component was implemented to manually assess the quality and accuracy of the LLM output content compared to the input snippets. Three domain experts with

extensive knowledge in construction delay analysis conducted detailed assessments of the generated summaries. Their reviews were conducted independently to remove bias and their evaluation used a standardized 1-5 scale rubric across four dimensions:

1. Event Description Accuracy

- 1: Multiple factual errors or misrepresentations
- 3: Minor inaccuracies present
- 5: Completely accurate representation of events

2. Date Accuracy

- 1: Multiple date errors or missing temporal information
- 3: Minor temporal inconsistencies
- 5: All dates accurately captured and presented

3. Event Capture Completeness

- 1: Missing multiple key events
- 3: Most important events captured with some omissions
- 5: Comprehensive coverage of all essential events

4. Language Quality

- 1: Unclear, poor grammar, or poorly structured
- 3: Generally clear with minor language issues
- 5: Clear, concise, and professionally written

C. Inter-rater Reliability

To ensure reliability and consistency between human evaluations, we employed Krippendorff's alpha (α) coefficient, a robust measure of inter-rater agreement. Krippendorff's alpha is calculated using the following formula:

$$\alpha = 1 - (Do/De)$$

where Do represents the observed disagreement among raters, and De represents the expected disagreement by chance. The coefficient ranges from -1 to 1, where 1 indicates perfect agreement, 0 suggests agreement equivalent to chance, and negative values indicate systematic disagreement. Following established research practices, this study set a minimum threshold of 0.667 for acceptable reliability. This threshold indicates substantial agreement among raters while acknowledging the inherent subjectivity in evaluating complex textual content.

The combination of automated metrics and expert human evaluation provided a robust framework for assessing both the

technical accuracy and practical utility of the generated summaries in the context of construction delay documentation. The evaluation framework incorporated both automatic metrics and human assessment to comprehensively evaluate model performance.

III. EXPERIMENTS

This section details the experimental setup for evaluating LLM performance in generating timeline summaries of construction delay events, following the methodology outlined in the previous section.

A. Dataset Preparation

The experimental data was synthesized through a systematic process designed to create realistic construction project communications while maintaining experimental control. Initially, ChatGPT generated synthetic construction project emails that simulated authentic project communications, capturing typical interactions, technical terminology, and delay-related discussions. These synthetic emails were then processed through a predictive modeling framework to identify documents containing delay-related content. Explainable predictive coding techniques were employed on the delay-related documents to extract snippets that have specific delay-related information. Twenty-one representative snippets encompassing various aspects of construction delays were selected for the summarization task. This methodical approach to data synthesis ensured both the controlled nature of our experiment and provided a reliable foundation for evaluating LLM timeline generation performance.

B. Experimental Setup

The following parameters and settings were used for all iterations to ensure consistency across results.

1) Model Configuration

The experiments used the Transformers text-generation pipeline with the following parameters optimized for summarization tasks:

- Temperature: 0.1 (prioritizing deterministic outputs)
- Maximum new tokens: 1,000
- Top-k: 30
- Top-p: 0.95 (nucleus sampling)
- No-repeat n-gram size: 4
- Repetition penalty: 1.2

These settings were chosen to (i) maintain consistency across outputs, (ii) minimize repetitive information, (iii) ensure comprehensive event coverage, and (iv) balance between conciseness and completeness.

2) Prompt Design

The experimental design incorporated a carefully crafted prompt to ensure structured and consistent outputs across all evaluated models. The prompt established a system role defining the LLM as an expert in construction project delay analysis,

providing a clear context for the task. Beyond role definition, the prompt specified precise output formatting requirements, including a three-section structure: an initial 100-word comprehensive summary of the delay situation, followed by a chronologically ordered list of dated events with source attributions, and finally, a collection of undated events with their corresponding source references. To maintain consistency and prevent hallucination in temporal information processing, the prompt included explicit rules prohibiting date inference when temporal information was not directly provided in the source text. Additionally, the prompt provided detailed formatting examples for both fully and partially dated events, ensuring standardized presentation of timeline information across all model outputs. The complete prompt structure is illustrated in Figure 1, demonstrating the framework designed to elicit consistent and comparable responses from the evaluated models.

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You are an expert in analyzing construction project delays. Your tasks are:

1. **Summary**:
- Provide a summary of the overall delay situation in **100 words**.

2. **Dated Events**:
- List events with date information in **chronological order**, including the source IDs for each event. Use these formats:
  a. For full dates (day, month, year):
  Example: 1. September 7, 2022 - Mobilization delayed due to high moisture, rain, and lack of matting. (Source IDs: 2, 3)
  b. For partial dates (e.g., no year):
  Example: 2. September 7 - Mobilization delayed due to weather conditions. (Source ID: 4)

3. **Undated Events**:
- For events without any date information, list them under a section titled "Undated Events", including the source IDs for each event. Use this format:
  Example: 1. Mobilization delayed due to high moisture, rain, and lack of matting. (Source IDs: 5, 6)

**Rules**:
- Do not infer dates or years if not explicitly provided.

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Figure 1 Prompt Used with Content and Format Requirements

3) Hardware Configuration

All experiments were conducted on a Microsoft Windows Server 2019 Datacenter virtual machine with dual AMD64 processors (2.45 GHz), 901GB RAM, and 934GB virtual memory. The system was equipped with four NVIDIA A100 80GB PCIe GPUs (total 320GB VRAM) running CUDA 12.1 with driver version 531.14, enabling efficient parallel processing for LLM inference. This standardized hardware configuration ensured consistent computational conditions across all model runs, crucial for accurate time cost comparisons.

IV. RESULTS

A. Automatic Metrics

We conducted five independent runs for each of the nine models (seven open-source LLMs and two commercial chatbots) to evaluate their performance in generating timeline summaries. The evaluation metrics included semantic similarity measures (BERTScore's precision, recall, and F1 Score) and lexical overlap metrics (ROUGE-1, ROUGE-2, and ROUGE-L scores). For open-source models, we also measured computational efficiency through CPU execution time. Table 1 presents the mean values of these metrics across all runs.

The results presented in Table 1 and visualized in Figures 2 and 3 reveal several key findings. In terms of overall performance, Claude showed superior results across all evaluation metrics, with both commercial chatbots (Claude and

ChatGPT) consistently outperforming open-source alternatives. Among all models evaluated, Phi-3-medium-128k-instruct showed the lowest performance compared to other models.

The comparison between semantic and lexical similarity metrics yielded interesting insights as shown in Figure 2. While ChatGPT achieved higher ROUGE scores than open source LLM results, indicating strong lexical overlap with the reference summary, the semantic similarity scores (BERTScore) were comparable across models. Particularly, Meta-Llama-3.1-70B-Instruct, Meta-Llama-3.1-8B-Instruct, and Mistral-7B-Instruct-v0.3 showed semantic understanding capabilities close to that of commercial solutions.

The computational efficiency analysis revealed significant variations in CPU execution time across models, primarily influenced by model size. Larger models exceeding 70B parameters, such as Meta-Llama-3.1-70B-Instruct, Mixtral-8x7B-Instruct-v0.1, and Qwen2-72B-Instruct, showed expected increased computation times due to their proximity to the hardware's computational capacity. Notably, among smaller models, Meta-Llama-3.1-8B-Instruct showed superior efficiency despite not having the smallest parameter count.

Analysis of performance stability across the five runs revealed distinct patterns. Commercial chatbots (Claude and ChatGPT) showed minimal variance, indicating highly stable performance. In contrast, open-source models demonstrated larger performance variations. Notable outliers in BERTScore performance were observed for Llama-3.1-70B-Instruct, Meta-Llama-3.1-8B-Instruct, Qwen2-72B-Instruct, Mixtral-8x7B-Instruct-v0.1, and Phi-3-medium-128k-instruct, where one round of results fell significantly below their normal performance range. However, Mistral-7B-Instruct-v0.3 and Gemma-2-9b-it maintained a more consistent BERTScore performance throughout all runs.

These comprehensive findings suggest that while commercial chatbots maintain a performance edge, several open-source alternatives offer comparable semantic understanding capabilities, albeit with greater performance variability. This performance variance requires careful consideration when selecting models for practical applications where consistency is crucial.

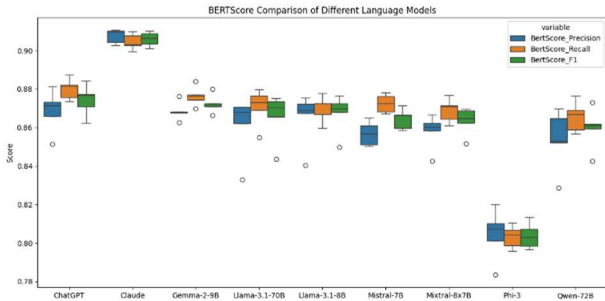


Figure 2 Semantic Similarity Comparison of LLM-Generated Summaries

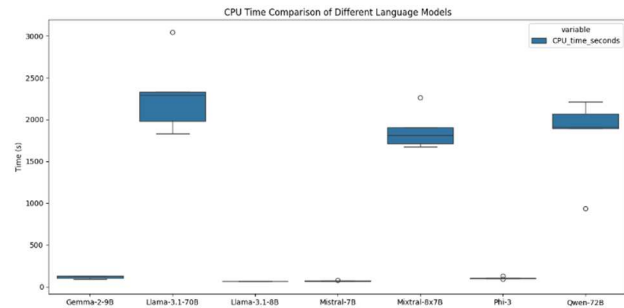


Figure 3 CPU Time Comparison of Different LLMs

B. Human Evaluation

While BERTScore and ROUGE scores offer valuable empirical insights into semantic similarity and lexical overlap, these scores cannot fully capture the generated summaries' crucial qualitative components. These aspects include adherence to prompt requirements, information accuracy, comprehensiveness of event extraction, temporal accuracy, source traceability, and linguistic quality. To address these limitations, we conducted a comprehensive human evaluation process across four critical dimensions: Event Description Accuracy, Date Accuracy, Event Capture Completeness, and Language Quality. To ensure evaluation consistency, each analyst independently reviewed all LLM-generated summaries using the standardized rubric described in the Methodology section.

The human evaluation results aligned with the empiric metrics results while offering additional qualitative insights. Claude showed superior performance across all four evaluation categories, followed closely by ChatGPT. Among open-source models, Llama-3.1-70B-Instruct exhibited the strongest adherence to prompt requirements and accuracy in event capture and source tracking. Qwen2-72B-Instruct showed promising overall performance, but demonstrated notable weaknesses in date extraction accuracy, performing better than only Phi-3-medium-128k-instruct in this aspect.

Temporal information processing emerged as a critical differentiator among models. Claude was the only model to achieve complete accuracy in date extraction and event association. Other models exhibited various temporal processing issues, including omission of date information, incorrect date-event associations, and, most concerning, the generation of fictitious dates, indicating hallucination tendencies in temporal reasoning.

Among smaller-scale models, Llama-3.1-8B-Instruct showed the most balanced performance, though it occasionally exhibited event omission or over-summarization tendencies. Commercial chatbots maintained superior linguistic quality standards, but all models produced acceptable summaries in terms of coherence and relevance. However, format adherence

varied, with Mixtral-8x7B-Instruct-v0.1 and Phi-3-medium-128k-instruct each notably displaying improper text segmentation in one iteration.

These human evaluation findings highlight the current capabilities and limitations of different LLMs in processing construction delay documentation, particularly emphasizing the challenges in accurate temporal information processing and consistent format adherence.

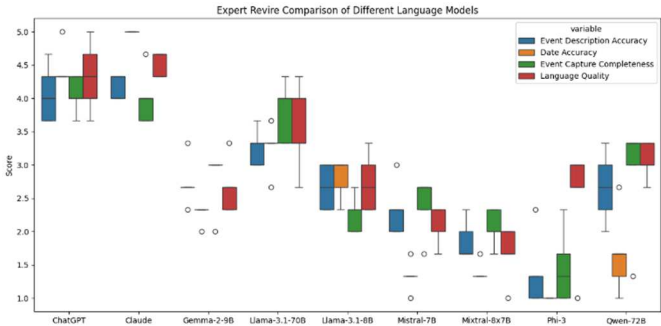


Figure 4 Expert Review Comparison of Different LLMs

V. CONCLUSIONS

This study presents a comprehensive comparative analysis of open-source LLMs generating timeline summaries of construction delay events. Our research addresses a critical need within construction litigation document review to create coherent, traceable summaries for critical delay events and leveraging various LLMs to achieve this goal.

We evaluated seven open-source LLMs and two commercial chatbots using carefully curated synthetic construction delay snippets. The evaluation framework incorporated both quantitative metrics and qualitative expert assessment. A gold-standard reference summary, created collaboratively by three experienced construction litigation analysts, served as the benchmark for metric calculations.

The results reveal several key findings with significant implications for practical applications. While Claude demonstrated superior performance across all evaluation metrics, most open-source LLMs, with the exception of Phi-3-medium-128k-instruct, showed comparable capabilities to commercial solutions. This finding suggests the viability of open-source alternatives for construction document analysis tasks. In terms of computational efficiency, smaller models, particularly Llama-3.1-8B-Instruct, showed impressive speed and offering practical advantages for resource-constrained applications.

Including human evaluation of the results provided crucial insights into model capabilities beyond statistical metrics. Commercial chatbots exhibited superior performance across all evaluation categories, with Claude being the only model to achieve complete accuracy in temporal information processing. Among open-source alternatives, Llama-3.1-70B-Instruct stood

out for its strong adherence to prompt requirements and accuracy in event capture and source tracking. For organizations seeking smaller-scale solutions, Llama-3.1-8B-Instruct offered the most balanced performance, though with occasional limitations in comprehensive event capture.

These findings have important implications for the construction industry. The ability of LLMs to generate accurate, traceable summaries of delay events can significantly streamline document review processes. Tracking the delay event sources enables reviewers to efficiently identify and access relevant detailed documentation. However, the varying performance in temporal processing and event capture accuracy suggests that human oversight remains essential, particularly for critical applications. Future work would include testing additional LLMs as they evolve and determining if substantial improvements can be made across all evaluation categories, human-reviewed and empirical.

Table 1 Comparison of ROUGE Scores, BERT Scores, and CPU Time for Different Models

Model	BertScore_Pre cision	BertScore_Re call	BertScore_F1	ROUGE1_F1	ROUGE2_F1	ROUGEL_F1	CPU time (s)
ChatGPT*	0.869	0.880	0.874	0.647	0.311	0.350	-
Claude*	0.908	0.905	0.906	0.699	0.450	0.460	-
Meta-Llama-3.1-70B-Instruct	0.861	0.871	0.866	0.495	0.153	0.230	2295
Meta-Llama-3.1-8B-Instruct	0.865	0.870	0.867	0.520	0.195	0.221	65
Mistral-7B-Instruct-v0.3	0.857	0.872	0.865	0.533	0.208	0.241	70
Mixtral-8x7B-Instruct-v0.1	0.858	0.869	0.863	0.485	0.192	0.217	1873
Phi-3-medium-128k-instruct	0.804	0.803	0.804	0.213	0.041	0.088	105
Qwen2-72B-Instruct	0.854	0.866	0.860	0.461	0.142	0.201	1804
gemma-2-9b-it	0.869	0.876	0.872	0.566	0.234	0.268	116

Table 2 Comparison of Expert Ratings for Different Models

Model	Event Description Accuracy		Date Accuracy		Event Capture Completeness		Language Quality	
	mean	std	mean	std	mean	std	mean	std
ChatGPT*	4.07	0.43	4.47	0.30	4.07	0.28	4.33	0.53
Claude*	4.20	0.18	5.00	0.00	4.00	0.41	4.53	0.18
Meta-Llama-3.1-70B-Instruct	3.27	0.28	3.27	0.37	3.80	0.45	3.53	0.65
Meta-Llama-3.1-8B-Instruct	2.67	0.33	2.80	0.30	2.27	0.28	2.73	0.43
Mistral-7B-Instruct-v0.3	2.33	0.41	1.33	0.24	2.40	0.43	2.13	0.30
Mixtral-8x7B-Instruct-v0.1	1.87	0.30	1.40	0.15	2.07	0.28	1.73	0.43
Phi-3-medium-128k-instruct	1.40	0.55	1.00	0.00	1.47	0.56	2.53	0.87
Qwen2-72B-Instruct	2.67	0.53	1.67	0.62	2.87	0.87	3.07	0.28
gemma-2-9b-it	2.73	0.37	2.27	0.15	2.80	0.45	2.67	0.41

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